

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (CSE/ISE/AIIML/CCE) (Credit System) Degree Examinations

April 2023

21CS301 / 21IS301 / 21AM301 / 21CC301 - STATISTICS AND PROBABILITY THEORY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) A discrete random variable X has the probability distribution as given below.

x_i	0	1	2	3	5	6	7
$P(x_i)$	0	K	2K	3K	K^2	$2K^2$	$7K^2+K$

Find K and evaluate $P[X < 6]$, $P[X \geq 6]$, and $P[0 < X < 5]$.

6 L*2 1 1
7 L2 1 2

- b) Find the mean and variance of binomial distribution.
c) The joint probability distribution of the random variables X and Y is given below.

Y \ X	-1	0	1
-1	0	0.1	0.1
0	0.2	0.2	0.2
1	0	0.1	0.1

- i) find marginal Pdf's of x and y ii) Are x and y independent?
iii) Find Cov(x,y).

7 L2 2 2

2. a) The pdf of a continuous random variable X is given as follows.

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

i) verify that f(x) is a pdf

ii) Evaluate $P[X \geq 1.5]$ and iii) Evaluate $P[1.5 \leq X \leq 2]$

6 L2 1 2
7 L2 1 2

- b) Find the mean and variance of exponential distribution.
c) The joint pdf of two continuous random variables X and Y is given by

$$f(x, y) = \begin{cases} x + y, & 0 \leq x, y \leq 1 \\ 0, & \text{elsewhere.} \end{cases}$$

Find E(X), E(Y), E(XY) and Cov(X,Y).

7 L3 2 2

3. a) Suppose that the random variable X has possible values 1, 2, 3, and $P[X=j] = \frac{1}{2^j}$, $j = 1, 2, 3, \dots$

i) compute $P[X \text{ is even}]$ ii) compute $P[X \geq 5]$

iii) compute $P[X \text{ is divisible by 3}]$

6 L2 2 2

- b) Suppose X is uniformly distributed over $(-a, a)$, where $a > 0$, whenever possible, determine 'a' so that the following are satisfied.

$$\text{i) } P[X > 1] = \frac{1}{3} \quad \text{ii) } P[X < \frac{1}{2}] = 0.7 \quad \text{iii) } P[|X| < 1] = P[|X| > 1]$$

7 L2 2 2

- c) State Bayes' theorem.

Mr. Sharma who lives in the outskirts of a city wants to catch an early morning train. The probability that he gets an auto at that time is only 0.23. If he gets an auto, the probability of catching the train is 0.85. If he fails to get an auto, he must take a city bus and the probability he gets train then is only 0.43. Find the probability that he catches the train.

7 L2 2 2

Unit – II

4. a) By the method of least square, fit a straight line to the following data.

x	1	2	3	4	5
y	14	13	9	5	2

6 L2 3 1

P.T.O.

- b) Find the correlation coefficient between x and y if

x	1	2	3	4	5
y	2	5	3	8	7

7 L2 3 1
7 L2 3 2

- c) State and prove Central limit theorem.

5. a) Fit a parabola to the following data.

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

6 L2 3 1

- b) Obtain the lines of regression for the following data.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

7 L2 3 2

- c) Given that, the sample mean and sample standard deviations of a random sample of size 150 are respectively 69.5 and 6.2. Find 99% confidence interval for the population mean.

7 L2 3 2

6. a) Calculate the rank correlation coefficients for the following data.

x	35	37	38	42	44	46	51	54	55	56
y	40	32	39	42	41	31	50	52	46	55

6 L2 3 2

- b) Given the regression lines,
- $4x - 5y + 33 = 0$
- and
- $20x - 9y - 107 = 0$
- .

i) Find the mean values of x and y.

ii) Find the correlation coefficient between x and y.

iii) Find the variance of y, if $V(X) = 9$.

7 L2 3 2

- c) Define moment generating function of a random variable, with an example. Show that the
- n^{th}
- derivative of m.g.f. of a random variable evaluated at
- $t=0$
- , yields
- $E[x^n]$
- .

7 L3 4 2

Unit – III

7. a) Define a stochastic process and a stochastic matrix. Verify that the

matrix $\begin{bmatrix} 0 & 0 & 1 \\ 0.5 & 0.25 & 0.25 \\ 0 & 1 & 0 \end{bmatrix}$ is a regular stochastic matrix.

6 L3 4 2

- b) Three boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. But C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball, find the probabilities that (i) A has the ball (ii) B has the ball and (iii) C has the ball, for the fourth throw.

7 L3 4 2

- c) Find the unique fixed probability vector of the following regular stochastic matrix

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Which matrix does P^n approach as n becomes larger?

7 L2 5 2

8. a) Find the unique fixed probability vector of the regular stochastic matrix

$$P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

6 L2 5 2

- b) Suppose that the probability of a dry day (state 0) following a rainy day (state 1) is
- $\frac{1}{3}$
- and that the probability of a rainy day following a dry day is
- $\frac{1}{2}$
- . Given that May 1 is a dry day, what is the probability that May 3 is a dry day and May 5 is a dry day.

7 L2 5 2

- c) Explain M|M|1 queuing.

7 L2 5 2